

5(a)	any three from: 1 small so can keep, in small space / for low cost / with low maintenance ; 2 can select parents for, mating / cross-breeding / particular crosses ; 3 so get many, offspring / generations, in a short time ; 4 can be used to research, cancer / (named) neurodegenerative diseases ; 5 AVP ;	3
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Question	Answer	Marks																									
5(b)	1 symbols: A = grey bodya = black body B = long wingb = short wing ; <i>F1 phenotypes:</i> grey long x grey long 2 <i>F1 genotypes:</i> AaBb x (AaBb) ; <table><tr><td></td><td>AB</td><td>Ab</td><td>aB</td><td>ab ;</td></tr><tr><td>AB</td><td>AABB</td><td>AABb</td><td>AaBB</td><td>AaBb</td></tr><tr><td>Ab</td><td>AABb</td><td>AAbb</td><td>AaBb</td><td>Aabb</td></tr><tr><td>aB</td><td>AaBB</td><td>AaBb</td><td>aaBB</td><td>aaBb</td></tr><tr><td>ab</td><td>AaBb</td><td>Aabb</td><td>aaBb</td><td>aabb ;</td></tr></table> 5 <i>F2 offspring phenotype number ratio:</i> 9 : 3 : 3 : 1 ; 6 grey long grey short black long black short ;		AB	Ab	aB	ab ;	AB	AABB	AABb	AaBB	AaBb	Ab	AABb	AAbb	AaBb	Aabb	aB	AaBB	AaBb	aaBB	aaBb	ab	AaBb	Aabb	aaBb	aabb ;	6
	AB	Ab	aB	ab ;																							
AB	AABB	AABb	AaBB	AaBb																							
Ab	AABb	AAbb	AaBb	Aabb																							
aB	AaBB	AaBb	aaBB	aaBb																							
ab	AaBb	Aabb	aaBb	aabb ;																							

Question	Answer	Marks
5(c)	1 carry out a test cross / cross with a double homozygous recessive fruit fly / cross with fly with black body and short wings ; <i>plus any two from:</i> 2 if some offspring have black body then F2 fly is, heterozygous for the body colour (gene) / Aa ; 3 if all offspring have grey body then F2 fly is, homozygous dominant for body colour (gene) / AA ; 4 if some offspring have short wings then F2 fly is, heterozygous for wing length (gene) / Bb ; 5 if all offspring have long wings then F2 fly is, homozygous dominant for wing length (gene) / BB ;	3

Question	Answer	Marks
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